

Alejandro Buendia

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Research Interests

Machine learning methods for genomic regulation, spatially-resolved data, and the adaptive immune response.

Education

Ph.D. Biomedical Data Science, Stanford University Sept. 2024 – present
Advisors: Anshul Kundaje and Scott D. Boyd.
B.A. Computer Science and Mathematics, Columbia University Feb. 2019
Undergraduate research advisor: Daniel Hsu.

Professional Experience

Columbia University Irving Medical Center, New York, NY Sept. 2023 – July 2024
Computational Research Lab Manager
Massachusetts Institute of Technology, Cambridge, MA Feb. 2022 – June 2023
Research Engineer / Junior Research Scientist, CSAIL
Microsoft, Cambridge, MA Jan. 2019 – Jan. 2022
Data Scientist II / Software Engineer

Publications

Preprints and Manuscripts Under Review

Chiho Im*, Liel Cohen-Lavi*, **Alejandro Buendia***, Anshul Kundaje, and Scott D. Boyd. BenchRep-T: a systematic evaluation of T-cell repertoire-based disease diagnostics. *bioRxiv*. (Author order determined randomly) 2026
Anvita Gupta, **Alejandro Buendia**, Anshul Kundaje, and Jure Leskovec. Bio-BLIP: a multimodal architecture for transferable reasoning in genomic variant interpretation. *bioRxiv*. 2026
Eric D. Sun*, **Alejandro Buendia***, Anne Brunet, and James Zou. SpatialProp: tissue perturbation modeling with spatially resolved single-cell transcriptomics. *bioRxiv*. 2025

Journal Articles

Samuel B. Hayward, Alina Vaitsiankova, Tomas Lama-Diaz, Juihsuan Chou, Angelo Tagliatela, Jen-Wei Huang, Yodharaudshani Wijesekarathanthi, Joshua R. Heyza, Giuseppe Leuzzi, Chuanyuan Chen, Nancy Wong, Tenzin Lhakhang, Xi Fu, **Alejandro Buendia**, Veronica Gheorghe, Nazanin Esmaeili Anvar, Jens C. Schmidt, Andre Nussenzweig, Raul Rabadan, Vincenzo Costanzo, Raphael Guerois, Traver Hart, and Alberto Ciccia. Mapping the genetic landscape of the DNA damage response with Cas12a-based combinatorial knockout screens. *Nature Biotechnology* (accepted in principle). 2026
Xi Fu*, Shentong Mo*, **Alejandro Buendia***, Anouchka Laurent, Anqi Shao, Maria del Mar Alvarez-Torres, Tianji Yu, Jimin Tan, Jiayu Su, Romella Sagatelian, Adolfo A. Ferrando, Alberto Ciccia, Yanyan Lan, David M. Owens, Teresa Palomero, Eric P. Xing, and Raul Rabadan. A foundation model of transcription across human cell types. *Nature*. 2025

Rebecca Boiarsky, Nalini M. Singh, **Alejandro Buendia**, Ava P. Amini, Gad Getz, and David Sontag. Deeper evaluation of a single-cell foundation model. *Nature Machine Intelligence*. 2024

Refereed Conference and Workshop Proceedings

- Rebecca Boiarsky, Nalini Singh, **Alejandro Buendia**, Gad Getz, and David Sontag. A deep dive into single-cell RNA sequencing foundation models. In *Machine Learning in Computational Biology (MLCB)*. Oral presentation (top 15%). 2023
- Stefan Hegselmann, **Alejandro Buendia**, Hunter Lang, Monica Agrawal, Xiaoyi Jiang, and David Sontag. TabLLM: few-shot classification of tabular data with large language models. In *Twenty-Sixth International Conference on Artificial Intelligence and Statistics (AISTATS)*. 2023
- Liqun Shao, Sahitya Mantravadi, Tom Manzini, **Alejandro Buendia**, Manon Knoertzer, Soundar Srinivasan, and Chris Quirk. Examination and extension of strategies for improving personalized language modeling via interpolation. In *First Workshop on Natural Language Interfaces (ACL)*. 2020
- Daniel Boley and **Alejandro Buendia**. Random walk fundamental tensor and graph importance measures. In *International Workshop on Big Social Media Data Management and Analysis (IJCAI)*. 2019
- Alejandro Buendia** and Daniel Boley. Optimized graph-based trust mechanisms using hitting times. In *International Workshop on Trust in Agent Societies (AAMAS)*. 2017

Teaching and Mentoring

Graduate Teaching Assistant, Foundation Models for Healthcare, Stanford University.

Undergraduate Teaching Assistant, Calculus IV and Discrete Mathematics, Columbia University.

Mentored one master’s thesis and two undergraduate researchers in clinical natural language processing and product deployment, MIT Clinical Machine Learning Group.

Service and Outreach

Editorial Board Member (current), *Genomics x AI Blog*. A cross-institution, open-source initiative publishing research findings and software for the genomics and AI communities.

Lead (current), qBMI (LGBTQ+ affinity group), Department of Biomedical Data Science, Stanford University.

Co-founder and lead, Diversity and Inclusion Initiative, Microsoft AI Development Program. Recruited underrepresented-minority graduates at academic and job-search conferences; partnered with Black in AI, LatinX in AI, and Women in ML.

Microsoft TEALS Outreach Program. Led a team of six developing a three-module introductory AI course for high school students; presented career talk about working at Microsoft to high school students in the Boston area.

Writing

Benchmarking AlphaGenome on NVIDIA GPUs: latency, memory, and feasibility across sequence lengths. *Genomics x AI Blog*. 2026

Porting AlphaGenome to PyTorch. *Genomics x AI Blog*. 2026

Fine-tuning AlphaGenome in native JAX/Haiku. *Genomics x AI Blog*. 2026

Software

`alphagenome_ft`: Fine-tuning utilities for AlphaGenome. Core developer.

`alphagenome-pytorch`: PyTorch implementation of AlphaGenome. Core developer.

Awards and Fellowships

Kluge Scholar, Columbia University.

Columbia College Summer Funding Program, Columbia University.